# Learning to Hear Motion Before Naming It

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## Abstract

An agent that *perceives* a continuous sensory stream should first have a representation of how the stream is changing before *naming* or *describing* it. This motivates the learning of latent motion systems, considering that movement can be learned through time and space from audio, among other sensors. We instantiate this work by proposing a two-stage approach on 8-channel raw audio of vehicles approaching an occluded junction. The first stage trains a predictor on the latent representation of raw multichannel waveforms with no labels. The second attaches a temporal model with three-task classification heads (direction, count, and type of vehicles being detected). With this approach, we investigate what self-supervised learning (SSL) contributes and what supervised learning changes. We find that the SSL representation captures the movement structure of the scene and the acoustic-source identity that distinguishes the vehicles from each other. The acoustic-vehicle data underlying these experiments will be released in a forthcoming multi-modal data collection paper.

## 1. Introduction

A growing line of work in representation learning shifts the prediction objective: rather than reconstructing pixels or waveforms, the model forecasts the latent representation of an unseen segment from context (LeCun & Courant, 2022). This change of target avoids spending capacity on perceptually irrelevant signal detail and has been instantiated for images (Assran et al., 2023), audio spectrograms (Tuncay et al., 2025), and video (Bardes et al., 2024). Recent work has begun to predict how scenes evolve between frames in videos (Garrido et al., 2026) to follow this principle of *understanding* the consequences of an action by being able to predict the next latent representation of the video. We extend this principle to its original temporal formulation (LeCun & Courant, 2022): from a past audio segment, the model forecasts the latent representation of a future one. We then distill this self-supervised representation into a temporal supervised classifier, so that acoustic motion is first *represented* and then *named* by direction, count, and type of vehicles, for a self-driving perception application.
This ordering is to echo how biological perception is organized. Object representation is built on the spatio-temporal regularities of *motion* (cohesion, continuity, contact) before it is organized by category (Spelke, 1990), and sensitivity to biomechanical motion appears in infants well before stable object naming (Bertenthal et al., 1984). Latent prediction is the mechanism for the first stage: what is predictable about a short future is precisely the motion structure of the scene. We compare our first results with supervised systems for direction detection of vehicles (Schulz et al., 2021; Li, 2023; Hao et al., 2024), which highlighted how sound can complement traditional sensors, and with a multi-task baseline proposed for traffic monitoring (Damiano et al., 2024). Working on a subset of a dataset we collected, we study left-to-right or right-to-left driving vehicles at a T-junction, and experiment with representation transfer to a back-to-front takeover scenario.

We aim for a controlled characterization of what each stage contributes: when does pre-training matter, what does supervised learning do to the representation, and how the SSL captures motion.

The contribution is built around three lines of evidence answering those questions:

- We present an architecture that beats the state-of-the-art models applied to our scope of application, working with a smaller dataset.
- We investigate the impact of SSL and supervised learning respectively.
- We demonstrate how SSL representation jointly organizes motion and source identity.

## 2. Method

We divide the description of the proposed framework into three parts: the temporal windowing of the input signal

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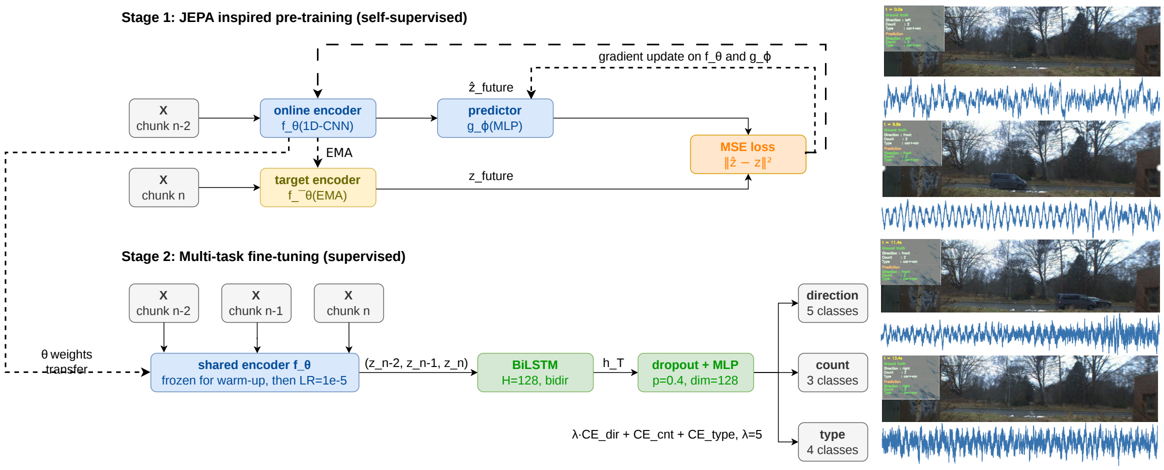



*Figure 1.* **Left:** the two-stage pipeline. Stage 1 trains a predictor on raw 8-channel audio to forecast the latent of a future second from its past context, with no labels. Stage 2 reuses the encoder under a BiLSTM with three task heads. **Right:** left-to-right car hidden by a van at a T-junction, with the model's prediction at a 1-second audio chunk.

(Section 2.1), the self-supervised pre-training of the acoustic encoder (Section 2.2), and the supervised multi-task fine-tuning stage (Section 2.3). An overview of the full pipeline is shown in Fig. 1.

### 2.1. Temporal Windowing of the Input Signal

Let the ego-vehicle carry a circular array of $M = 8$ microphones sampled at $f_s = 48\,\text{kHz}$. The continuous multi-channel data point is segmented into non-overlapping one-second audio frames that we call chunks. Each chunk is represented as a raw waveform tensor $\mathbf{x} \in \mathbb{R}^{M\times L}$, where $L = f_s = 48K$ samples. Each chunk is normalized channel-wise to zero mean and unit variance, with no spectral transform: the model operates directly on the raw waveform.

Consecutive chunks are grouped into sliding windows of length $T = 3$ seconds. The window centred at chunk $n$ is the ordered triple

$$\mathbf{W}_n = \big(\mathbf{x}_{n-2},\ \mathbf{x}_{n-1},\ \mathbf{x}_n\big), \tag{1}$$

where subscripts denote absolute chunk indices within a fixed data point scenario. Windows do not cross the scenario boundaries.

### 2.2. Self-Supervised Pre-training

#### 2.2.1. Architecture

The acoustic encoder $f_\theta : \mathbb{R}^{M\times L} \to \mathbb{R}^D$ maps a single one-second waveform to a latent vector of dimension $D = 256$. It consists of three strided one-dimensional convolutional blocks followed by global average pooling and a linear projection:

$$\mathbf{z} = \mathbf{W}_{\text{fc}}\ \text{GAP}\Big(\phi_3\big(\phi_2\big(\phi_1(\mathbf{x})\big)\big)\Big), \tag{2}$$

where $\mathbf{x}$ denotes the normalized input, each block $\phi_k$ is a one-dimensional strided convolution with kernel size 9 and stride 2, followed by batch normalization and a ReLU activation, producing $\{64, 128, 256\}$ output channels for $k \in \{1, 2, 3\}$ respectively. GAP denotes global average pooling over the time dimension, and $\mathbf{W}_{\text{fc}} \in \mathbb{R}^{D\times 256}$ is the linear projection matrix. The choice $D = 256$ matches the channel depth of the final convolutional block, so no information bottleneck is introduced before the latent space.

A lightweight predictor $g_\phi : \mathbb{R}^D \to \mathbb{R}^D$ approximates the mapping from a past latent representation to a future one. It is a two-layer MLP:

$$g_\phi(\mathbf{z}) = \text{Linear}_{\phi_2}\Big(\text{ReLU}\big(\text{Linear}_{\phi_1}(\mathbf{z})\big)\Big), \tag{3}$$

where $\text{Linear}_{\phi_k}$ denotes a fully-connected layer with learnable parameters $\phi_k$, both mapping $\mathbb{R}^D \to \mathbb{R}^D$.

#### 2.2.2. Training Objective

Following the JEPA principle (LeCun & Courant, 2022; Assran et al., 2023), we maintain two encoder instances: an *online* encoder $f_\theta$ updated by gradient descent, and a *target* encoder $f_{\bar\theta}$ updated by an exponential moving average (EMA) of the online weights:

$$\bar\theta\ \leftarrow\ m\,\bar\theta + (1-m)\,\theta, \quad m = 0.99. \tag{4}$$

with momentum *m* following the convention established by BYOL (Grill et al., 2020) and adopted by I-JEPA (Assran et al., 2023). The target encoder receives no gradient; it provides stable prediction targets that prevent representational collapse without requiring negative pairs (Grill et al., 2020). The target encoder is updated by EMA after every gradient step on $\theta$.

For a window $\mathbf{W}_n$, the online encoder processes the *past* chunk $\mathbf{x}_{n-2}$ and the predictor is trained to forecast the target encoding of the *future* chunk $\mathbf{x}_n$:

$$\mathcal{L}_{\text{SSL}} = \left\| g_\phi\big(f_\theta(\mathbf{x}_{n-2})\big) \; - \; f_{\bar\theta}(\mathbf{x}_n) \right\|_2^2. \tag{5}$$

The intermediate chunk $\mathbf{x}_{n-1}$ is intentionally excluded from pre-training, forcing the encoder to capture longer-range acoustic dynamics (see section 3 for ablation). Gradients flow only through $f_\theta$ and $g_\phi$; the target encoder $f_{\bar\theta}$ is updated solely via Eq. (4). Pre-training runs for 100 epochs on the training split using the Adam optimizer (Kingma & Ba, 2015) ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, batch size 32). After pre-training, the online encoder weights $\theta$ are retained and the predictor $g_\phi$ is discarded.

### 2.3. Supervised Multi-Task Fine-tuning

#### 2.3.1. Model architecture

The pre-trained encoder $f_\theta$ is composed with a bidirectional LSTM and three classification heads (Fig. 1). The three chunks of window $\mathbf{W}_n$ are encoded independently by the shared encoder, producing the latent sequence $\big(\mathbf{z}_{n-2}, \mathbf{z}_{n-1}, \mathbf{z}_n\big) \in \mathbb{R}^{T\times D}$ with $\mathbf{z}_{n+k} = f_\theta(\mathbf{x}_{n+k})$, $k \in \{-2, -1, 0\}$. A single-layer bidirectional LSTM with per-direction hidden dimension $H = 128$ aggregates the sequence; at each step the forward and backward hidden states are concatenated as $\overleftrightarrow{\mathbf{h}}_{n+k} = [\overrightarrow{\mathbf{h}}_{n+k}; \overleftarrow{\mathbf{h}}_{n+k}] \in \mathbb{R}^{2H}$, with $2H = D = 256$ so the BiLSTM output lives in the same ambient dimension as the encoder latent. Only the final step $\overleftrightarrow{\mathbf{h}}_n$ is propagated forward, through a shared module $\mathbf{h} = \mathrm{MLP}_\psi(\overleftrightarrow{\mathbf{h}}_n) \in \mathbb{R}^{128}$ consisting of a linear projection $\mathbb{R}^{2H} \to \mathbb{R}^{128}$ with a ReLU non-linearity, regularized by dropout with probability p=0.4 applied to both its input and output. Three linear heads $\mathbf{W}_{\text{dir}} \in \mathbb{R}^{5\times128}$, $\mathbf{W}_{\text{cnt}} \in \mathbb{R}^{3\times128}$, $\mathbf{W}_{\text{type}} \in \mathbb{R}^{4\times128}$ project $\mathbf{h}$ to logits for the five direction classes (*left, front, right, none, both/back*), three count classes (0, 1, 2), and four vehicle-type classes (*car, van, car + van, none*).

#### 2.3.2. Training objective and protocol

Each head is trained with cross-entropy; the total loss is

$$\mathcal{L} = \lambda\,\mathcal{L}_{\text{CE}}^{\text{dir}} + \mathcal{L}_{\text{CE}}^{\text{cnt}} + \mathcal{L}_{\text{CE}}^{\text{type}}, \quad \lambda = 5, \tag{6}$$

with the direction term up-weighted because it is the most environment-sensitive of the three (Schulz et al., 2021). Fine-tuning proceeds in two phases. During a warm-up of $E_w = 5$ epochs, $f_\theta$ is frozen and only $\mathrm{MLP}_\psi$ and the heads are trained with Adam ($\eta_{\text{head}} = 10^{-3}$). Afterwards, $f_\theta$ is unfrozen and jointly optimized at $\eta_{\text{enc}} = 10^{-5}$, the rate is chosen empirically to adapt the SSL representation without catastrophic forgetting. Total training runs for $E = 20$ epochs at batch size 32; the checkpoint with the highest validation direction accuracy is retained.

*Table 1.* Direction-task baselines. Validation accuracy on the static *T-junction* set. All baselines re-implemented on our recordings.

| **Method** | **Direction (%)** |
|---|---|
| SRP-PHAT + SVM (Schulz et al., 2021) | 53.4 |
| DOA + TF + CNN (Hao et al., 2024) | 61.9 |
| Mel-spec. + CNN (Li, 2023) | 75.0 |
| SSL + BiLSTM (ours) | **81.8** |

## 3. Experiments

After presenting the dataset briefly (3.1), we test the ordering claim in two complementary ways: an ablation that isolates the contribution of pre-training (3.2) and a representation-level comparison of the encoder (3.3) before diving into what the representation actually contains (3.4).

### 3.1. Dataset

We use a real-world dataset of 9,800 labeled chunks across six recording scenarios and four acoustic environments with 6-class taxonomy for the Direction task, the class *both* is in the static T-Junction set, and the unseen class *back* is in the driving take-over set.

### 3.2. The contribution of pre-training

We compare performances on existing supervised techniques. Table 1 contrasts our pipeline with prior single-task NLOS direction methods re-implemented on our data (Schulz et al., 2021; Li, 2023; Hao et al., 2024). Table 2 presents the results of supervised-only multi-task baselines: the plain BiLSTM trained from scratch, and the DCASE 2024 task 10 challenge baseline (Damiano et al., 2024). We note that for a dataset of this size, the supervised classifiers alone cannot give results as good as with our implementation; which suggests that the representation is learned at the SSL stage.

### 3.3. Representation-level analysis of the encoder

Table 2 varies encoder initialization (random vs. ours) and training mode (frozen encoder vs. fine-tuning) maintaining everything else in the pipeline and parameters held identical. With the same temporal probe (BiLSTM), our SSL over random frozen encoder adds 20.51%, 18.06%, 22.3% accuracy improvement for the direction, count, and type tasks respectively. Finetuning adds a slight improvement with a random encoder (from 4.52% up to 6.18%), and a marginal one for our SSL encoder (from 0.4% up to 2.37%), which indicate the encoder captures mainly the representation of the motion and its acoustical identity.

*Table 2.* Multi-task classification on the *T-junction* validation set, reporting accuracy (%). The *prediction-target* ablation compares the proposed two-step horizon, against the *interpolation*, and the *one-step* variants. *Take-over scenario*: the SSL encoder $f_\theta$ is frozen and only the BiLSTM, the shared MLP, and the three classification heads are fine-tuned on the held-out take-over set; each cell reports mean $\pm$ std over 10-shot adaptation.

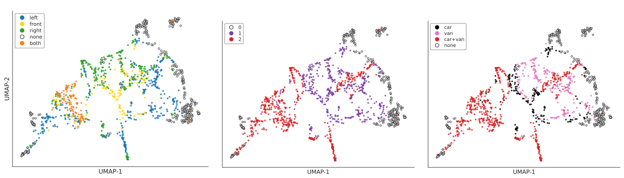
| **Method / Config.** | **Direction** | **Count** | **Type** |
|---|---|---|---|
| *Baselines & Encoder initialization* | | | |
| DCASE (Damiano et al., 2024) | 67.5 | 89.5 | 66.9 |
| BiLSTM only | 33.7 | 47.0 | 40.0 |
| Random + BiLSTM (frozen) | 66.3 | 79.7 | 76.2 |
| Random + BiLSTM (fine-tuned) | 70.4 | 83.3 | 80.3 |
| SSL + BiLSTM (frozen) | 79.9 | 94.1 | 93.2 |
| SSL + BiLSTM (fine-tuned - ours) | **81.8** | **94.5** | **93.7** |
| *Prediction target* | | | |
| Interpolation | 79.3 | 89.0 | 87.3 |
| One-step | 80.8 | 94.1 | 92.5 |
| *Take-over scenario* | | | |
| ours (10-shot) | 70.1±10.1 | 85.7±3.1 | 56.8±3.6 |

The prediction-target ablation compares three configurations of the same encoder-predictor architecture, differing only in which chunks act as context and target. The *one-step* variant predicts $\mathbf{x}_n$ from $\mathbf{x}_{n-1}$; the *interpolation* variant recovers $\mathbf{x}_{n-1}$ from $\{\mathbf{x}_{n-2}, \mathbf{x}_n\}$. The two-step horizon performs best. One possible interpretation: one-step prediction can be largely achieved by extrapolating short-time content from the immediately preceding chunk, providing a weaker inductive bias than the longer horizon; conversely, predicting the masked intermediate chunk from a bidirectional context turns the pretext task closer to reconstruction than to forecasting. We investigated how the representation transfers across recording configurations. On a held-out driving (take-over) dataset, the accuracy reaches 56.8% for the type classification task, 85.7% for the counting, 70.1% for the direction task with only ten labeled recordings used to extend the head (*back* class added). The significant drop in the accuracy of the type classification task reflects the distribution shift between stationary and moving ego-vehicle conditions. However, the limited drops for the other two tasks suggest that the learned representation is not specific for each task it was trained on.

### 3.4. How motion carries identity

The pretext task asks for temporal predictability of a latent: $g_\varphi$ minimizes $\|g_\varphi(f_\theta(x_{n-2})) - f_{\bar{\theta}}(x_n)\|_2^2$ without seeing any label. However, with the intuition that the temporal evolution of an agent's acoustic signature is not separable from what that agent is (*e.g*: a van is slow low-frequency engine modulating over an approach is both its motion signature and its identity), we analyze the SSL latent abilities to capture jointly the motion and the source identity, and ask what the representation actually contains without supervision.

Figure 2 shows a UMAP projection of the learned latents (fit once and colored by each task classes) as a qualitative experiment to the metrics reported above. Three structural facts are visible. First, silence consistently lives outside the active manifold: the *none* class of direction, the *0* class of count, and the *none* class of type occupy the same peripheral islands, indicating a stable *no vehicle* state separated from any vehicle activity. Second, the dominant axis within the active manifold separates one-vehicle from two-vehicle scenarios. Figure 2-b shows a clean spatial gradient with two-vehicle chunks (*2*) on the left and one-vehicle chunks (*1*) on the right. Figure 2-c inherits this structure since two-vehicle scenarios are always car+van pairs by recording protocol. Third, within the one-vehicle subset of 2-c, *car* is less clustered than *van*, demonstrating a weaker representation of this class. Figure 2-a shows that direction-of-arrival classes form recognizable regions: *left* (blue) and *right* (green) occupy opposite sides of the active manifold, *both* (orange) clusters tightly on the left edge where the two-vehicle region lies, and *front* (yellow) intermixes with the others, partially in coherence with the two opposite sides, but also highlighting the encoder's limits to capture the spatial structure of that class.



(a) (b) (c)

*Figure 2.* UMAP projection of the frozen-encoder validation latents (same projection coordinates across all three panels), coloured by the three downstream classification labels. (a) Direction (b) Count (c) Type.

## 4. Conclusion and Future Work

We have shown that, on raw multichannel audio of an occluded-junction perception task, a JEPA-style pretext with a two-step latent-prediction horizon produces a representation that already organizes both the motion of the scene and the acoustic identity of its sources, and that a small temporal probe reads this representation out to multi-task accuracy above existing supervised pipelines trained on substantially larger labeled dataset. The supervised stage adds essentially nothing the pre-training has not already provided, consistent with an ordering in which an agent first *represents* the dynamics of a sensory stream and only then *names* what it has organized. A natural next step is to re-use the SSL predictor $g_\varphi$ at inference time, quantifying how far each future is from what the encoder expected, and turning the present passive-classification system into one that reasons about its own anticipations to be more robust when facing unknown situations.

## 5. Acknowledgement

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